Saiteja Vulli

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Education

Rowan University — Glassboro, NJ
Masters in Data Science — *Jan 2025 – Dec 2026*

SR University — Warangal, TS
Bachelor's in Electrical and Electronics Engineering — *Jun 2020 – May 2024*

Experience

Graduate Research Assistant (Deep Learning & Generative AI) — Jan 2026 – Present
Rowan University — *Glassboro, NJ*

- Conducted research on diffusion models and representation learning.
- Developed VQ-VAE and diffusion pipelines using PyTorch and MONAI.
- Integrated transformers and CLIP embeddings for structured-to-latent modeling.
- Optimized models via GPU training, hyperparameter tuning, and evaluation.
- Collaborated with faculty to refine architectures and improve model performance.

Frontend Developer — Jan 2023 – Dec 2024
Aagama IT Biz Solutions Pvt Ltd — *Hyderabad, Telangana*

- Developed UI components for a real estate web application using Node.js, improving user experience.
- Integrated APIs for real-time data updates and seamless functionality.
- Built responsive interfaces using CSS, Bootstrap, and JavaScript.
- Optimized performance by reducing re-renders and improving load times.

Projects

AetherFlow - AI Research Data Platform | *Python, Apache Kafka, PySpark, Spark, Docker, Parquet, Streamlit*

- Built an end-to-end real-time data pipeline for AI/ML experiment telemetry using Apache Kafka and Spark Structured Streaming with a Bronze–Silver–Gold Medallion architecture.
- Developed PySpark ETL pipelines and Parquet analytical datasets for model performance, experiment tracking, and GPU utilization, with an interactive Streamlit dashboard for monitoring and analysis.

Banking Fraud Detection Data Platform | *Python, Apache Kafka, Apache Spark, PostgreSQL, Docker, Streamlit*

- Built an end-to-end real-time data engineering pipeline to simulate, process and analyze banking transactions for fraud detection using Apache Kafka and Spark Structured Streaming.
- Developed rule-based fraud detection, transaction validation and Dead Letter Queue (DLQ) handling, storing processed transactions, fraud alerts and invalid records in PostgreSQL.
- Created analytical SQL views and interactive dashboards using Streamlit/Tableau to monitor fraud trends, transaction volumes, fraud rates and high-risk transaction patterns.
- Containerized the complete platform using Docker and Docker Compose, demonstrating event-driven architecture, real-time streaming, SQL analytics and scalable data engineering practices.

Parquet Compactor CLI | *Rust, Polars, Clap, Indicatif, TOML*

- Developed a high-performance Rust CLI tool to solve the small file problem by compacting multiple Parquet files into optimized larger files.
- Implemented schema validation, configurable target file sizes, dry-run mode, verbose logging and TOML-based configuration support.
- Built efficient file discovery, batching and Parquet processing using Polars, with progress tracking via Indicatif and robust error handling.
- Added multi-threading support, Criterion benchmarks and detailed summary reporting to improve processing efficiency, scalability and reliability.

Resume Analyzer | *Java, Spring Boot, React, PostgreSQL*

- Built a full-stack Resume Analyzer for PDF resume upload and job matching.
- Developed REST APIs using Spring Boot for resume upload, text extraction, and match score analysis.
- Integrated PostgreSQL for storing resume data and used Apache PDFBox to extract text from PDF files.
- Designed a responsive React frontend to display match scores, matched skills, missing skills, and suggestions.

NYC Yellow Taxi Data Analysis | *Rust, DataFusion*

- Built a data analytics pipeline in Rust using Apache DataFusion to process large-scale NYC taxi data (Parquet).
- Performed aggregations (monthly revenue, trip trends, tip analysis) using DataFrame and SQL APIs.
- Optimized data processing by loading and querying multiple files as a unified dataset.
- Generated insights on trip patterns and payment behavior through efficient query execution.

Crack Detection using TensorFlow | *Python,Google Colab, Git*

- Developed a CNN-based binary image classification model in TensorFlow to detect surface cracks.
- Applied data preprocessing and model optimization techniques to improve accuracy.
- Automated training and evaluation using Python, leveraging accuracy and confusion matrix metrics.

Technical Skills

Languages: Java, Python, Rust, SQL (PostgreSQL), JavaScript, C/C++, HTML/CSS, R
Frameworks & Libraries: Spring Boot, React.js, Node.js, FastAPI, Flask, Polars, Pandas, NumPy, Matplotlib, Streamlit
Databases & Messaging: PostgreSQL, MongoDB, Apache Kafka
Big Data & Analytics: Apache Spark Structured Streaming, Apache Parquet, Tableau
Developer Tools: Git, GitHub, Docker, Docker Compose, pgAdmin, Google Cloud Platform, IntelliJ IDEA, VS Code, PyCharm, Visual Studio, MATLAB
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs, ETL Pipelines, Event-Driven Architecture, Real-Time Data Processing, Multithreading, CLI Development, Performance Optimization, Benchmarking

Certifications

- Convolutional Neural Networks – Coursera
- Data Analysis with Python – IBM
- SQL and Relational Databases – IBM